# Temperature, Heat and 1st Law of Thermodynamics

Temperature 0th Law of Thermodynamics

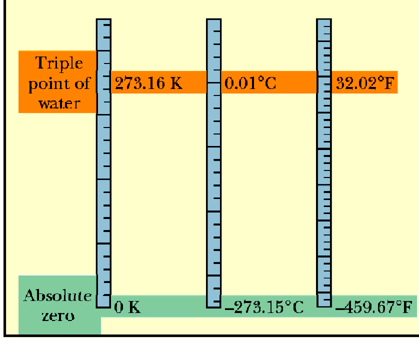
Thermal expansion Heat and Work

1st Law of Thermodynamics

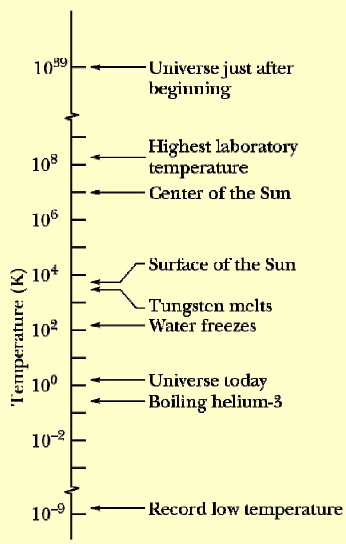
# Temperature

- SI unit: Kelvin – K
- Other units: $T_C = (T_K - 273.15)$

$$T_F = \frac{9}{5}T_C + 32°$$




Lower limit

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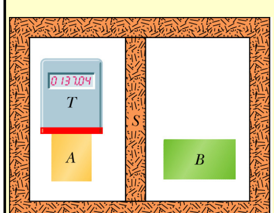
# Problem

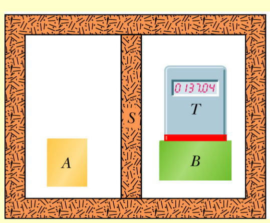
- A body is initially at a temperature of 293 K and then the temperature increases to 313 K.
  - Convert these temperatures to °C.
  - Determine the difference in the two temperatures in K and °C.

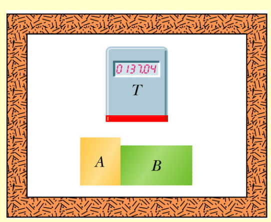


# $0^{th}$ Law of Thermodynamics

- Properties of many materials change with temperature:
  - Liquid increases in volume with increase in temp.
  - Metal rod increases in length with increase in temp.
  - Electrical resistance of a wire increases with increase in temp.
  - Pressure of enclosed gas increases with increase in temp.
- $0^{th}$ Law of Thermodynamics: If body A and body B are each in thermal equilibrium with a third body *T*, then *A* and *B* are in equilibrium with each other.

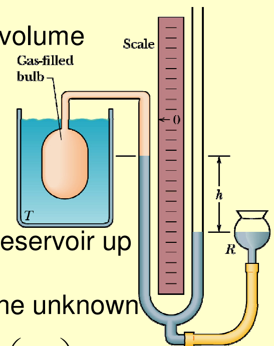


# Measuring Temperature

- Triple point of water
  - Standard fixed point with a fixed temperature
  - $T_3 = 273.16$ K
- Constant-Volume gas thermometer
  - Based on the pressure of a gas in a fixed volume
  - Fill a thermometer bulb with some gas
  - Attach it to a Hg manometer
  - Measure the pressure $p_3$ when bulb is at triple point: $p = p_0 - \rho g h$
  - $T_3 = C p_3$
  - Always keep the volume constant (move reservoir up and down)
  - Measure the pressure $p$ when bulb is at the unknown temperature: $p = p_0 - \rho g h$
  - Eliminating unknown $C$: $T = Cp$ , $T = T_3 \left( \frac{p}{p_3} \right)$

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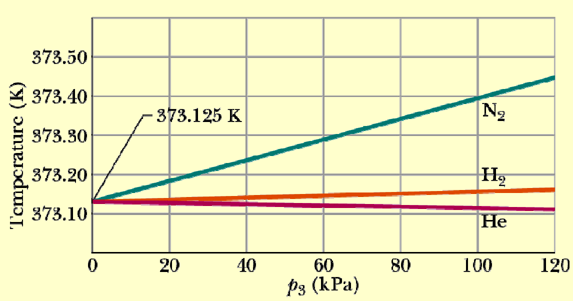
# Measuring Temperature

- $T = T_3 \left( \frac{p}{p_3} \right) = 273.16 \left( \frac{p}{p_3} \right)$
- Different results for $H_2$, He and $N_2$, depending on pressure. As $p \to 0$ the measured temperature converges for the various gases.

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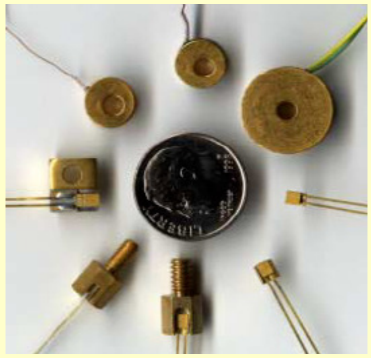
# Thermal properties

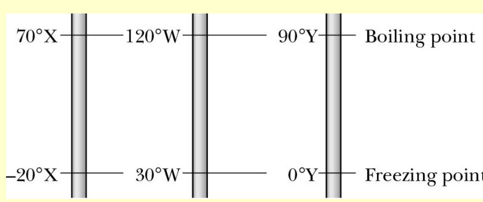
- Thermocouples





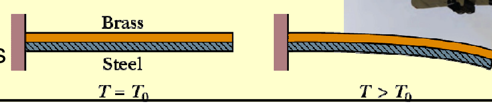
# Checkpoint 1

- The figure shows 3 linear temperature scales with freezing and boiling points of water indicated. (a) Rank the degrees on these scales by size. (b) Rank the following temperatures highest first: 50 °X, 50°W and 50°Y.



# Thermal Expansion

- What happens when the temperature of a material increases?
  - Energy is added
  - Atoms move a bit further apart
  - Different materials' atoms move different distances further apart
- Thermal expansion must be taken into account in everyday situations or used to our advantage.
  - Bridge, train tracks and buildings have expansion joints
  - Dental fillings
  - Concorde (due to frictional forces)
  - Bimetal strip
  - Thermostats
  - Thermometers

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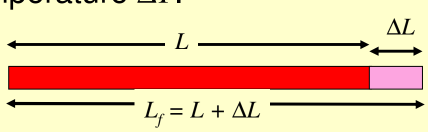
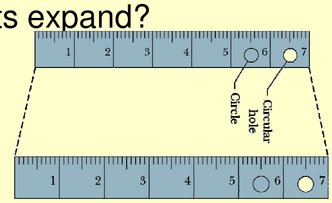
# Linear Expansion

- Consider a metal rod of length $L$ which is raised by a temperature $\Delta T$.




- Its length will increase from $L$ to $L_f$ : $\Delta L = L\alpha\Delta T$
- $\alpha$ is the coefficient of linear expansion
- Units: $°C^{-1}$ or $K^{-1}$
- How do holes and voids in objects expand?
  - Holes
  - Circle
  - Scale
  - Numbers

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## Linear Expansion

TABLE 18-2

Some Coefficients of Linear Expansion[a]

| Substance | $\alpha(10^{-6}/C°)$ | Substance | $\alpha(10^{-6}/C°)$ |
|---|---|---|---|
| Ice (at 0°C) | 51 | Steel | 11 |
| Lead | 29 | Glass (ordinary) | 9 |
| Aluminum | 23 | Glass (Pyrex) | 3.2 |
| Brass | 19 | Diamond | 1.2 |
| Copper | 17 | Invar[b] | 0.7 |
| Concrete | 12 | Fused quartz | 0.5 |



## Volume Expansion

- All dimensions increase with an increase in temperature.
- Volume must also increase.
- $\Delta V = V\beta\Delta T$
- β is the coefficient of volume expansion
- $\beta = 3\alpha$

## Why does ice float in water?

- Above 4 ℃ water expands with an increase in temperature like other solids – density decreases.
- Between 0 to 4 ℃ water contracts with increase in temperature.
- At 4 ℃ water has a max. density.

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## Problem

- A steel train track has a length of 30 m when the temperature is 32 ℉. What is its length when the temperature is (a) 313 K? (b) -40℃?

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# Sample Problem p483

- On a hot day in LA, an oil trucker loaded 37 000 L of diesel fuel. He encountered cold weather on his journey, and where it was 23.0 K lower than in LA, he delivers his entire load. How many litres did he deliver? The coefficient of volume expansion for diesel is $9.50 \times 10^{-4}/C°$, and the coefficient of linear expansion for the steel tank is $11 \times 10^{-6}/C°$.

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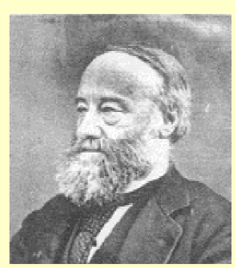
# Temperature and Heat

- Thermal energy is an internal energy that consists of the kinetic and potential energies of the random motion of the atoms and molecules in the object.
- Heat $Q$: is the energy transferred between a system and its environment because of a temperature difference between them.
- Heat is a form of energy – measured in Joule
- Other common units:
  - 1 calorie (cal) = 4.1860 J
  - 1 Calorie (Cal) = 1kcal = 4 186.0 J [Diet]



J.P. Joule 1818 – 1889

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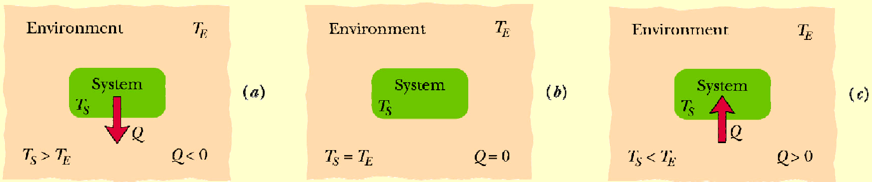
# Temperature and Heat

- $T_S > T_E$ $\quad$ $T_S = T_E$ $\quad$ $T_S < T_E$
- $Q < 0$ $\quad$ $Q = 0$ $\quad$ $Q > 0$




**Sign convention:**
Flow out of system: $Q < 0$
Flow into system: $Q > 0$

**Internal Energy** of a system CAN change without energy being transferred by heat. e.g.
- Compress a gas with a piston – no heat flow, but temperature and internal energy increases.
- When a gas expands rapidly – it cools.

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# Absorption of Heat –Solids and liquids

- Heat capacity $C$ – proportionality constant (J/K) between $Q$ and $\Delta T$

$$Q = C\Delta T = C\left(T_f - T_i\right)$$

- Specific Heat (Capacity) $c$ – Heat capacity per unit mass (J/kg·K)

$$Q = cm\Delta T = cm\left(T_f - T_i\right)$$

- Heat of transformation $L$ – Heat absorbed by the sample when it changes phase – temperature is constant (J kg$^{-1}$)

$$Q = Lm$$

  - $L_F$ heat of fusion (freezing or melting)
  - $L_V$ heat of vaporisation (boiling or condensing)

Table 18-3 and 18-4

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# Absorption of Heat

TABLE 18-3

Some Specific Heats and Molar Specific Heats at Room Temperature

| Substance | Specific Heat $\frac{\text{cal}}{\text{g}\cdot\text{K}}$ | Specific Heat $\frac{\text{J}}{\text{kg}\cdot\text{K}}$ | Molar Specific Heat $\frac{\text{J}}{\text{mol}\cdot\text{K}}$ |
|---|---|---|---|
| *Elemental Solids* | | | |
| Lead | 0.0305 | 128 | 26.5 |
| Tungsten | 0.0321 | 134 | 24.8 |
| Silver | 0.0564 | 236 | 25.5 |
| Copper | 0.0923 | 386 | 24.5 |
| Aluminum | 0.215 | 900 | 24.4 |
| *Other Solids* | | | |
| Brass | 0.092 | 380 | |
| Granite | 0.19 | 790 | |
| Glass | 0.20 | 840 | |
| Ice ($-10°C$) | 0.530 | 2220 | |
| *Liquids* | | | |
| Mercury | 0.033 | 140 | |
| Ethyl alcohol | 0.58 | 2430 | |
| Seawater | 0.93 | 3900 | |
| Water | 1.00 | 4180 | |

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# Absorption of Heat

TABLE 18-4

Some Heats of Transformation

| Substance | Melting: Melting Point (K) | Melting: Heat of Fusion $L_F$ (kJ/kg) | Boiling: Boiling Point (K) | Boiling: Heat of Vaporization $L_V$ (kJ/kg) |
|---|---|---|---|---|
| Hydrogen | 14.0 | 58.0 | 20.3 | 455 |
| Oxygen | 54.8 | 13.9 | 90.2 | 213 |
| Mercury | 234 | 11.4 | 630 | 296 |
| Water | 273 | 333 | 373 | 2256 |
| Lead | 601 | 23.2 | 2017 | 858 |
| Silver | 1235 | 105 | 2323 | 2336 |
| Copper | 1356 | 207 | 2868 | 4730 |

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## Checkpoint 3

- A certain amount of heat $Q$ will warm 1g of material $A$ by 3 C° and 1 g of material $B$ by 4 C°. Which material has the greater specific heat?

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## Sample Problem p488

- How much heat must be absorbed by ice of mass $m$ = 720 g at -10 ℃ to take it to liquid state at 15℃?

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# Sample Problem p488

- If we supply the ice with a total of only 210 kJ (as heat), what are the final temperature and state of the water?:

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# Sample Problem p 487

- A copper slug whose mass $m_c$ is 75 g is heated in a lab oven to 312℃. The slug is then dropped into a glass beaker containing a mass $m_w$ = 220 g of water. The heat capacity $C_b$ of the beaker is 45 cal/K .The initial temperature $T_i$ of the water and the beaker is 12 ℃. Assuming the slug, beaker and the water are an isolated system and the water does not vaporize, find the final temperature $T_f$ of the system at thermal equilibrium.

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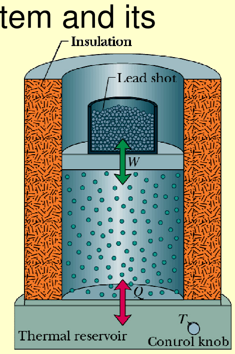
# Sample Problem p487



# Heat and Work

- Energy can be transferred between system and its environment as heat and work.
  - Gas confined to cylinder with moving piston
  - Upward force on piston due to gas pressure = weight of load on piston.
  - Walls are insulated – no heat transfer
  - Bottom of cylinder is on a thermal reservoir temperature $T$ (controllable)



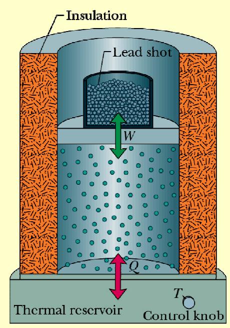

- Initial State $i$: $p_i$, $V_i$, $T_i$
- Thermodynamic process results in change to
- Final State $f$: $p_f$, $V_f$, $T_f$

# Heat and Work

- Thermodynamic process could be:
  - Transfer of energy from reservoir into system: +ve heat
  - Transfer of energy out of the system into reservoir : −ve heat
  - Work can be done by system – raise the piston: +ve work
  - Work can be done on system – lower the piston: -ve work
- Quasi-static: Slow change so that the system is always in thermal equilibrium.
- Work done by the system:

$$dW = \vec{F}\cdot d\vec{s} \qquad p = \frac{F}{A}$$
$$= (pA)(ds)$$
$$= pV \qquad V = Ads$$
$$W = \int dW = \int_{V_i}^{V_f} pV$$

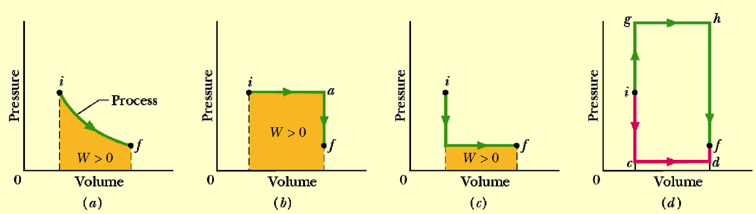
  - Area under the $pV$ curve



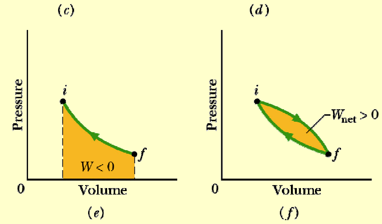



# Heat and Work

- Many ways for system to get from state $i$ to state $f$.

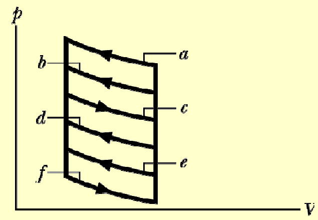



- Sign convention:
  - Gas does work: $W > 0$
  - Work done on gas: $W < 0$
- Changing of state ($i$ to $f$) by $W$ or $Q$.
- $i$ to $f$ : $W \neq Q$ and path-dependent



# Checkpoint 4

- The *p*-*V* diagram here shows 6 curved paths (connected by vertical paths) that can be followed by a gas. Which 2 of the curved paths should be part of a closed cycle (those curved paths plus connecting vertical paths) if the net work done by the gas during the cycle is to be at its maximum positive value?



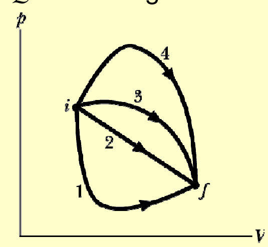



# 1$^{st}$ Law of Thermodynamics

- When changing state from $i$ to $f$, $Q$ *and* $W$ depend on the nature of the process.
- But $Q$ - $W$ is same for all processes – path independent.
- $Q - W = \Delta E_{\text{int}} = E_{\text{int},f} - E_{\text{int},i}$
- First Law of Thermodynamics
- $dE_{\text{int}} = dQ - dW$
- The internal energy $E_{\text{int}}$ of a system tends to increase if energy is added as heat $Q$ and tends to decrease if energy is lost as work $W$ done by the system.
- Ch 8 – Isolated system – no energy transferred into or out of the system.

# Checkpoint 5

- The figure shows 4 paths on the $p$-$V$ diagram along which a gas can be taken from state $i$ to $f$. Rank the paths according to (a) the change $\Delta E_{int}$ of the gas, (b) the work done by the gas, and (c) the magnitude of the energy transferred as heat $Q$ between gas and the environment, greatest first.

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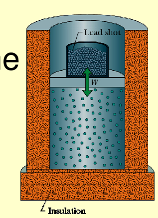
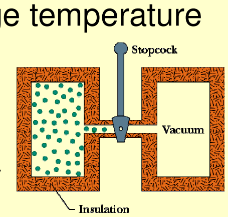
# Special cases of the 1st Law




- Adiabatic Process: $Q = 0$, $\Delta E_{int} = -W$ i.e. if the system does work ($W$ is +ve) the change in internal energy is negative.
- Isovolumetric Process: $W = 0$, $\Delta E_{int} = Q$ i.e. $\Delta V = 0$, heat is absorbed or lost.
- Cyclical Process: $W = Q$, $\Delta E_{int} = 0$ i.e. the system is restored to the initial state $i$, so no change temperature and internal energy is unchanged.
- Free expansions: $W = 0$, $Q = 0$, $\Delta E_{int} = 0$ i.e. adiabatic process where no heat is transferred and no work is done on or by the system.

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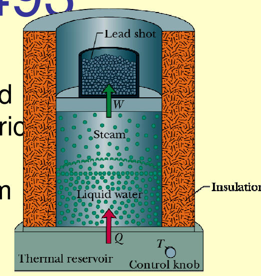
# Sample Problem p493




- Let 1.00 kg of liquid water be at 100℃ be converted to steam at 100℃ by boiling at standard atmospheric pressure (1.00 atm = $1.01\times10^5$ Pa) in an arrangement shown. The volume of the water changes from $1.00\times10^{-3}$ m$^3$ as a liquid to 1.671 m$^3$ as steam.
  (a) How much work is done by the system during this process?
  (b) How much energy is transferred as heat during the process?
  (c) What is the change in the system's internal energy?

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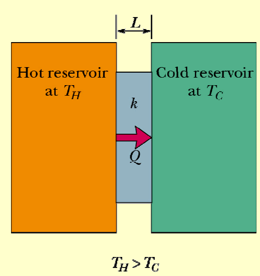
# Heat Transfer Mechanisms

- Thermal conduction
  – Metal spoon left in the pot when cooking




- Convection
  – Heater in the room
- Radiation
  – Solar energy

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